

Crossing the Steel River

I can match a river-crossing problem to two different steel bridge solutions.

[Open the original interactive lesson](#)

Arrival · retrieve the previous move

Standard

Order 1911, then 1934.

Pupil-chosen arrival variant

Arrival rule: retrieve before adding new content. Personal or family disclosure is never required.

Starter · question before verdict

Enquiry

How did engineers help people cross a busy working River Tees?

Retrieval question: In this evidence-informed constructed design problem, what difficulty could a low fixed road create?

- A. It could block taller river traffic
- B. River water was too cold
- C. Bridges were banned

I Do · model the disciplinary move

Model the decision and evidence, not just the answer.

STEP Need: carry people and vehicles across.

STEP Evidence-informed constructed constraint: provide a crossing while allowing river clearance; this is a classroom design brief, not a quotation from the engineers.

STEP 1911 Transporter Bridge: a high frame supports a moving gondola.

Bridge matcher · date, movement, reason

I Do · model the disciplinary move

Choose a bridge, then the card that matches it. Diagrams are simplified, not to scale.

I Do · model the disciplinary move

- A. Transporter Bridge
- B. Newport Bridge
- C. opened 1911
- D. opened 1934
- E. a gondola hangs from a high frame and glides across
- F. the whole road deck lifts up between two towers
- G. ships pass under the frame while the gondola waits
- H. traffic stops while the deck is raised for a ship

I Do · model the disciplinary move

Pick a bridge.

A. What can the diagram NOT show?

We Do · evidence hinge

Commit before reveal: Which Transporter feature kept a road deck out of the channel?

- A. A suspended moving gondola
- B. Blue paint
- C. A fixed low roadway

I Do 2 · connect and qualify

Model connection, limitation and honest uncertainty.

STEP The 1931–34 Newport Bridge used a liftable central road span.

STEP The broad problem was similar; the moving parts were different.

STEP Compare date, need, mechanism and evidence—not looks.

ADAPTED SOURCE CARD

HUM-B-W13-C01 · Transporter Bridge: stable facts

Provenance source: HUM-B-W13-S01

I Do 2 · connect and qualify

Middlesbrough Council states that the Tees Transporter Bridge between Middlesbrough and Port Clarence opened in 1911. Its gondola was intended to carry vehicles and pedestrians across the River Tees.

Can establish Establish opening year, location and intended gondola function.

Cannot establish Establish current operating status or every reason for choosing the design.

Accessible route Three labels: 1911; high frame; suspended gondola carries people and vehicles across.

HUM-B-W13-C02 · Newport Bridge: stable facts

I Do 2 · connect and qualify

Provenance source: HUM-B-W13-S03

Historic England records Newport Bridge as built in 1931–34 with a vertically lifting centre road span. When raised, it provided 36 metres of clearance for river traffic at high water.

Can establish Establish date, vertical-lift mechanism and recorded clearance.

Cannot establish Establish current lifting capability or every reason for selecting this design.

Accessible route Two-frame description: road span down for crossing; centre span raised for 36 m high-water clearance.

I Do 2 · connect and qualify

CONSTRUCTED TEACHING SOURCE — created for classroom practice. It is not an archival original.

HUM-B-W13-C03 · Constructed comparison brief

Provenance source: HUM-B-W13-S02

Classroom brief: people and vehicles need to cross, while river traffic may need clearance. Compare how each moving mechanism could respond. This brief is an evidence-informed inference, not a quotation from the engineers.

Can establish Support structured comparison of the documented mechanisms.

I Do 2 · connect and qualify

Cannot establish Prove the complete historical decision process behind either bridge.

Accessible route Need → possible constraint → mechanism: cross the river → allow clearance → gondola across or centre span up.

We Do 2 · interactive humanities lab

Interactive probe: response → evidence or representation → adult receives → next question or scaffold changes.

Bridge Constraint Lab

CONSTRUCTED TEACHING SOURCE — created for classroom practice. It is not an archival original.

Disciplinary focus: Constraint → engineering solution and comparison

Classify the evidence, operate each constructed mechanism model, then use the evidence-informed classroom brief to explain need → possible constraint → solution.

We Do 2 · interactive humanities lab

TRANSPORTER

- 1911
- suspended gondola
- Middlesbrough–Port Clarence

NEWPORT

- built 1931–34
- lifting central road span
- 36 m high-water clearance when raised

We Do 2 · interactive humanities lab

BOTH

- steel river crossing
- moving designs that could provide river clearance

Transporter

- A. Gondola at Middlesbrough side
- B. Gondola travelling beneath high frame
- C. Gondola at Port Clarence side

We Do 2 · interactive humanities lab

Newport

- A. Centre road span down for road crossing
- B. Centre road span rising
- C. Centre road span raised with 36 m high-water clearance

We Do 2 · interactive humanities lab

Evidence classification

- A. TRANSPORTER 1911
- B. TRANSPORTER suspended gondola
- C. TRANSPORTER Middlesbrough–Port Clarence
- D. NEWPORT built 1931–34
- E. NEWPORT lifting central road span
- F. NEWPORT 36 m clearance when raised
- G. BOTH steel river crossing

We Do 2 · interactive humanities lab

H. BOTH moving clearance solution

Accessible route: Transporter: high frame and suspended carriage. Newport: road span shown down and raised. Text describes both movements; arrows and labels supplement colour.

Independent · build defensible evidence

Independent humanities mission

A dated bridge comparison using need, constraint, mechanism and one evidence limitation.

16 minutes. Supported, Standard and Stretch are movable routes, not pupil labels.

HUMANITIES STUDIO READY

Source / materials Teacher-prepared constructed lab cards plus the locally approved source cards named for this lesson.

Independent · build defensible evidence

Care / access Use the stated text, visual, pointing, sorting, speaking or exactly scribed route. No personal or family disclosure is required.

Evidence target A dated bridge comparison using need, constraint, mechanism and one evidence limitation.

Supported

Match six bridge cards.

Evidence: Matched board plus one because-link.

The ____ Bridge used ____.

Independent · build defensible evidence

Fade the first prompt no longer needed.

Standard

Complete a need, constraint and solution matrix for both bridges.

Evidence: Comparison matrix.

People needed ____, but ships needed ____, so ____.

Stretch

Judge how each solved its period's problem and add a source limit.

Evidence: Comparative judgement.

Independent · build defensible evidence

Both _____. Unlike _____. The evidence cannot tell _____.

Exit · receive, interpret, influence

Standard

Explain one problem–solution link.

VOICE: The pupil compares one bridge problem–solution link through the selected exit route and chooses INDUSTRY FIRST or NATURE FIRST for Week 14.

One BUILD Humanities close only; no signature, receipt, upload queue or second closure form.